Official Hugging Face notebook: resnet-50
Notebook run in Google Colab, with its saved outputs. Opened from: https://huggingface.co/microsoft/resnet-50.ipynb

In [1]:
```
!pip install -U transformers
```

```
Requirement already satisfied: transformers in
/usr/local/lib/python3.13/dist-packages (5.20.2)
Requirement already satisfied: huggingface-hub<3.0,>=1.31.0 in
/usr/local/lib/python3.13/dist-packages (from transformers) (1.33.0)
Requirement already satisfied: numpy>=1.17 in
/usr/local/lib/python3.13/dist-packages (from transformers) (2.1.3)
Requirement already satisfied: packaging>=20.0 in
/usr/local/lib/python3.13/dist-packages (from transformers) (26.3)
Requirement already satisfied: pyyaml>=5.1 in
/usr/local/lib/python3.13/dist-packages (from transformers) (6.0.3)
Requirement already satisfied: regex>=2025.10.22 in
/usr/local/lib/python3.13/dist-packages (from transformers) (2025.11.3)
Requirement already satisfied: tokenizers<0.24.0,>=0.23.1 in
/usr/local/lib/python3.13/dist-packages (from transformers) (0.23.3)
Requirement already satisfied: typer in /usr/local/lib/python3.13/dist-
packages (from transformers) (0.27.3)
Requirement already satisfied: safetensors>=0.8.0 in
/usr/local/lib/python3.13/dist-packages (from transformers) (0.8.0)
Requirement already satisfied: tqdm>=4.60 in
/usr/local/lib/python3.13/dist-packages (from transformers) (4.67.3)
Requirement already satisfied: click<9.0.0,>=8.4.2 in
/usr/local/lib/python3.13/dist-packages (from huggingface-
hub<3.0,>=1.31.0->transformers) (8.5.0)
Requirement already satisfied: filelock>=3.10.0 in
/usr/local/lib/python3.13/dist-packages (from huggingface-
hub<3.0,>=1.31.0->transformers) (4.2.4)
Requirement already satisfied: fsspec>=2023.5.0 in
/usr/local/lib/python3.13/dist-packages (from huggingface-
hub<3.0,>=1.31.0->transformers) (2025.12.0)
Requirement already satisfied: hf-xet<2.0.0,>=1.6.0 in
/usr/local/lib/python3.13/dist-packages (from huggingface-
hub<3.0,>=1.31.0->transformers) (1.7.0)
Requirement already satisfied: httpx<1,>=0.23.0 in
/usr/local/lib/python3.13/dist-packages (from huggingface-
hub<3.0,>=1.31.0->transformers) (0.28.1)
Requirement already satisfied: typing-extensions>=4.1.0 in
/usr/local/lib/python3.13/dist-packages (from huggingface-
hub<3.0,>=1.31.0->transformers) (4.16.0)
Requirement already satisfied: anyio in /usr/local/lib/python3.13/dist-
packages (from httpx<1,>=0.23.0->huggingface-hub<3.0,>=1.31.0-
>transformers) (4.15.1)
Requirement already satisfied: certifi in /usr/local/lib/python3.13/dist-
packages (from httpx<1,>=0.23.0->huggingface-hub<3.0,>=1.31.0-
>transformers) (2026.10.19)
Requirement already satisfied: httpcore==1.* in
/usr/local/lib/python3.13/dist-packages (from httpx<1,>=0.23.0-
>huggingface-hub<3.0,>=1.31.0->transformers) (1.0.9)
Requirement already satisfied: idna in /usr/local/lib/python3.13/dist-
packages (from httpx<1,>=0.23.0->huggingface-hub<3.0,>=1.31.0-
>transformers) (3.20)
Requirement already satisfied: h11>=0.16 in
/usr/local/lib/python3.13/dist-packages (from httpcore==1.*-
```

```
>httpx<1,>=0.23.0->huggingface-hub<3.0,>=1.31.0->transformers) (0.16.0)
Requirement already satisfied: shellingham>=1.3.0 in
/usr/local/lib/python3.13/dist-packages (from typer->transformers) (1.5.4)
Requirement already satisfied: rich>=13.8.0 in
/usr/local/lib/python3.13/dist-packages (from typer->transformers)
(13.9.4)
Requirement already satisfied: annotated-doc>=0.0.2 in
/usr/local/lib/python3.13/dist-packages (from typer->transformers) (0.0.5)
Requirement already satisfied: markdown-it-py>=2.2.0 in
/usr/local/lib/python3.13/dist-packages (from rich>=13.8.0->typer-
>transformers) (4.2.0)
Requirement already satisfied: pygments<3.0.0,>=2.13.0 in
/usr/local/lib/python3.13/dist-packages (from rich>=13.8.0->typer-
>transformers) (2.21.0)
Requirement already satisfied: mdurl~=0.1 in
/usr/local/lib/python3.13/dist-packages (from markdown-it-py>=2.2.0-
>rich>=13.8.0->typer->transformers) (0.1.2)
```

Local Inference on GPU

Model page: https://huggingface.co/microsoft/resnet-50

⚠️ If the generated code snippets do not work, please open an issue on either the model repo and/or on huggingface.js 🙏

In [2]:
```python
# Use a pipeline as a high-level helper
from transformers import pipeline

pipe = pipeline("image-classification", model="microsoft/resnet-50")
pipe("https://huggingface.co/datasets/huggingface/documentation-
images/resolve/main/hub/parrots.png")
```

```
Warning: You are sending unauthenticated requests to the HF Hub. Please
set a HF_TOKEN to enable higher rate limits and faster downloads.
WARNING:huggingface_hub.utils._http:Warning: You are sending
unauthenticated requests to the HF Hub. Please set a HF_TOKEN to enable
higher rate limits and faster downloads.
```

config.json: 100% 69.6k/69.6k [00:00<00:00, 2.04MB/s]

model.safetensors: reconstructing file: 100% 102MB / 102MB, 9.42MB/s

model.safetensors: downloading bytes: 96.1MB, 8.28MB/s

Loading weights: 100% 320/320 [00:00<00:00, 1664.55it/s]

preprocessor_config.json: 100% 266/266 [00:00<00:00, 10.4kB/s]

Out[2]:
```
[{'label': 'macaw', 'score': 0.9999986886978149},
 {'label': 'vulture', 'score': 2.1576383346655348e-07},
 {'label': 'sulphur-crested cockatoo, Kakatoe galerita, Cacatua
galerita',
  'score': 2.3804894055956538e-08},
 {'label': 'cocktail shaker', 'score': 1.8397100731704086e-08},
 {'label': 'toucan', 'score': 1.1318117110192816e-08}]
```

In [3]:
```python
# pip install -U transformers accelerate
# Load model directly
from transformers import AutoImageProcessor,
```

```
AutoModelForImageClassification

processor = AutoImageProcessor.from_pretrained("microsoft/resnet-50")
model =
AutoModelForImageClassification.from_pretrained("microsoft/resnet-50",
device_map="auto")
```

Loading weights: 100% 320/320 [00:00<00:00, 1558.61it/s]

Remote Inference via Inference Providers

Ensure you have a valid **HF_TOKEN** set in your environment. You can get your token from your settings page. Note: running this may incur charges above the free tier. The following Python example shows how to run the model remotely on HF Inference Providers, automatically selecting an available inference provider for you. For more information on how to use the Inference Providers, please refer to our documentation and guides.

In [ ]:
```
import os
os.environ['HF_TOKEN'] = 'YOUR_TOKEN_HERE'
```

In [ ]:
```
import os
from huggingface_hub import InferenceClient

client = InferenceClient(
    provider="auto",
    api_key=os.environ["HF_TOKEN"],
)

output = client.image_classification("cats.jpg",
model="microsoft/resnet-50")
```